Fisica
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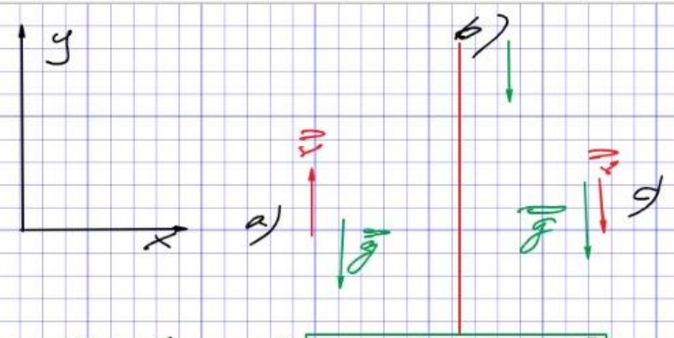
Todos los marcadores



Vector:
Es una cantidad matemática q' tiene magnitud dirección y sentido

posición
velocidad
aceleración, fuerza.

Cual es la aceleración?
a) b) c) la aceleración es la gravedad (Tierra) y siempre apunta hacia abajo

a) velocidad hacia arriba (sube)
aceleración hacia abajo (gravedad)
velac. ↑ acel. ↓ el cuerpo desacelera (baja la velac.)

c) velocidad hacia abajo (baja)
aceleración hacia abajo (gravedad)
velac. ↓ acel. ↓ el cuerpo acelera.
(aumenta la velocidad)

displac... (variación de posición)

speed (rapidez: es magnitud de la velac.)

a) La aceleración de una roca mientras sube es igual a 9,8 m/s² apunta hacia abajo
vel. sube

b) En el punto más alto la aceleración es igual 9,8 m/s² apunta hacia abajo
velac cero

La aceleración de una roca mientras baja es igual a 9,8 m/s² apunta abajo
vel. baja

b) En el cenit (punto máximo) la velocidad es cero.
la aceleración va hacia abajo

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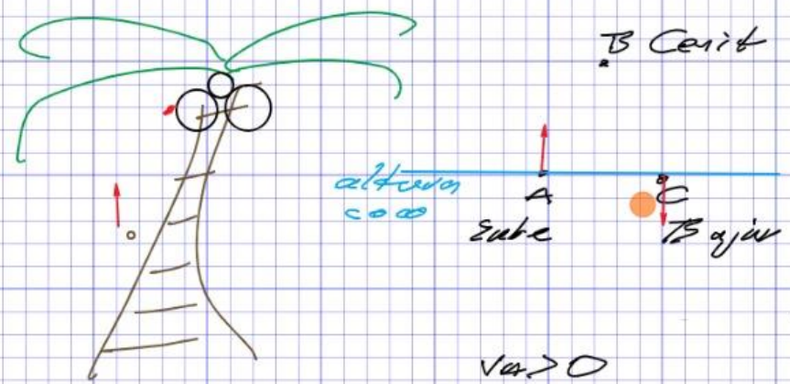
Δ : cambio



$v_A > 0$

$v_B = 0$

$v_C < 0$

desplazamiento

$$v_f^2 = v_0^2 + 2a(\Delta d)$$

$$v_f^2 = v_0^2 + 2ad$$

$A \to B$

$$\cancel{v_B^2} = v_A^2 - 2gd$$

$$0 = v_A^2 - 2gd$$

$$2gd = v_A^2$$

$B \to C$

$$v_C^2 = \cancel{v_B^2} + 2gd$$

$$v_C^2 = 2gd$$

$$v_C^2 = 2gd$$

Igualo

$$\boxed{v_A^2 = v_C^2}$$

$v_A = v_C$ la velocidad de la roca en A es igual a en C.

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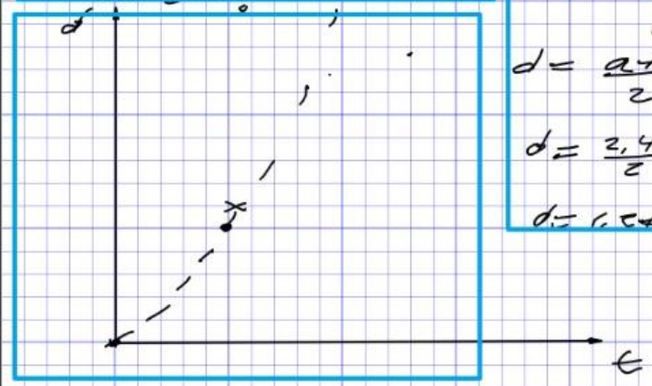
52. $v_i = 0$ (reposo)

Datos $\begin{cases} a = 2{,}4\ m/s^2 \\ t = 12\ s \end{cases}$

Incog: $d = ?$



cte

pos. inicial

$d = \cancel{d_i} + \cancel{v_i t} + \frac{at^2}{2}$

velocc. inicial

$d = \frac{at^2}{2}$

$d = \frac{2{,}4}{2} t^2$

$d = 1{,}2t^2$

t	d
0	0
1	1,2
2	4,8

$v_f = \cancel{v_i} + at$

$v_f = 2{,}4 \cdot 12$

$\boxed{v_f = 28{,}8\ m/s}$

$t = 12\ s \Rightarrow d = 1{,}2 \cdot 12^2$

$d = 1{,}2 \cdot 144$

$\boxed{d = 172{,}8\ m}$

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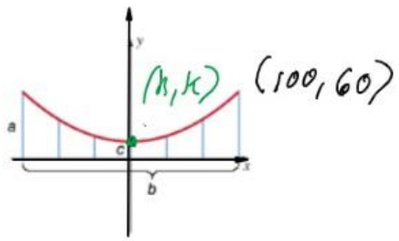
axes : ejes

Solve the problem.

One section of a suspension bridge hangs between two towers that are a = 60 feet above the surface and b = 200 feet apart, as shown in the figure below. A cable strung between the tops of the two towers is in the shape of a parabola with its vertex c = 40 feet above the surface. With axes drawn as indicated in the figure, find the equation of the parabola.

$\frac{1}{500}x^2 + 40$



Repaso

$(a+b)^2 = (a+b)(a+b)$
$= aa + ab + ba + b \cdot b$
$= a^2 + ab + ab + b^2$
$= a^2 + 2ab + b^2$

$(x+3y)^2 = x^2 + 2 \cdot x \cdot 3y + (3y)^2$
$= x^2 + 6xy + 9y^2$

Desarrollo

(1)

Vértice (h,k)

$k=c$: distancia vertical entre el vértice y el origen de ordenadas.

$h=0$: " horizontal " " " " " " " "

Ecuación Parábola (Orientación vertical)

$(y-k) = p(x-h)^2$

$p>0$ Abierta arriba
$p<0$ " abajo

$(y-c) = p(x-0)^2$

$(y-40) = p(x)^2 \Rightarrow (y-40) = px^2$

(2)

$(y-40) = px^2$

$(60-40) = p \cdot 100^2$

$20 = 10000p$

$p = \frac{20}{10000}$

$p = 0{,}002$

(3)

$y - 40 = 0{,}002x^2$

$y = 40 + 0{,}002x^2$

$1000y = 40000 + 2x^2$

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Find the vertex, focus, and directrix of the given parabola.

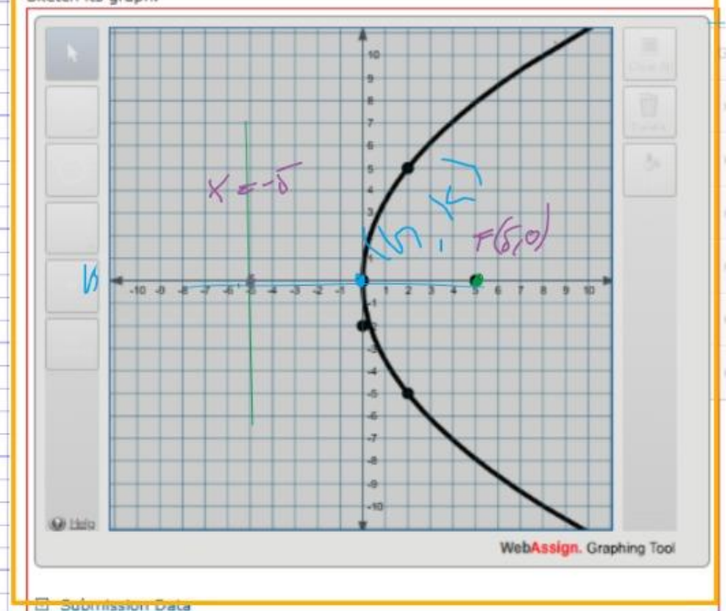
$y^2 = 20x$

vertex $(x, y) = (0, 0)$

focus $(x, y) = (5, 0)$

directrix -5

Sketch its graph.



Submission Data

1

h : coord. en x

k : " en y

$(y-k) = p(x-h)^2 \longleftrightarrow$ vertical

$(x-h) = p(y-k)^2 \longleftrightarrow$ horizontal

2

$y^2 = 20x$ Horizontal (El término y está al cuadrado)

3

$20x = y^2$

$x = \frac{1}{20} y^2$

$x = x - h \Longrightarrow \boxed{h=0} \Rightarrow x = x$

$\frac{1}{20} y^2 = p(y-k)^2 \Rightarrow \boxed{k=0} \Rightarrow \frac{y^2}{20} = py^2 \Rightarrow \boxed{\frac{1}{20} = p}$

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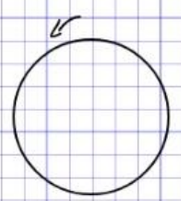
Todos los marcadores

Exercise 1: The Ceiling Fan (Frequency & Velocity)

- **Scenario:** A ceiling fan blade (radius $r = 0.6$ m) rotates at 300 RPM (revolutions per minute).
- **Tasks:**
  1. Convert RPM to Frequency (f in Hz) and Period (T in seconds).
  2. Calculate angular velocity (ω in rad/s).
  3. Calculate the linear speed (v) of the tip of the blade.
  4. **Challenge:** What is the linear speed of a point halfway to the center ($r = 0.3$ m)?
- **Key Connection:** $v = r\omega$ (Speed depends on radius); $\omega = 2\pi f$.

$f = 300/60 = \frac{30}{6} = 5\ Hz$

$T = \frac{1}{f} = \frac{1}{5} = 0{,}2\ sec$

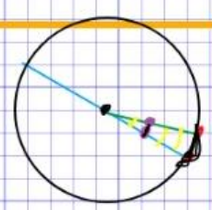


10 vueltas 1000 sec.

$T = ?$ $\frac{1000}{10} = 100\ (sec.)$

$f = \frac{1}{T} = \frac{1}{100} = 0{,}01\ (sec^{-1})\ (Hz)$

$T = \frac{1}{f}$



angular velocity: ángulo recorrido en cierta cantidad de tiempo (rad/s) $\omega = \frac{\Delta\theta}{\Delta t}$

linear speed: distancia recorrida en cierta cantidad de tiempo (m/s) $v = \frac{\Delta s}{\Delta t}$

Period: cantidad de tiempo q' tarda (s) en dar una vuelta

Frequency: # vueltas q' se ejecutan en cierta cantidad de tiempo (Hz)

θ: theta : ángulo recorrido (rad)
t: tiempo (s)
s: longitud de arco (m)
Δ: delta (cambio)

$\Delta s = s_f - s_i$ $\Delta x = x_f - x_i$ $\Delta v = v_f - v_i$

$x_i - \Delta x - x_f$

$\omega = \frac{\Delta\theta}{\Delta t} = \frac{\theta_f - \theta_i}{t_f - t_i} = \frac{\pi/2 - 0}{25 - 0}$

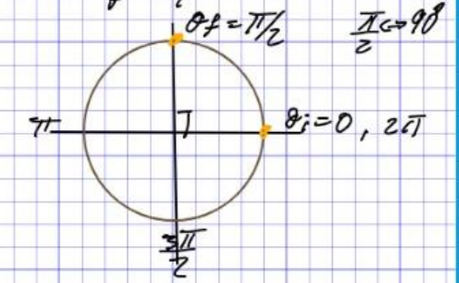
$\omega = \frac{\pi/2}{25} = \frac{\pi/2}{25/1} = \frac{\pi}{50}$

$\omega = 0{,}0628\ rad/s$

$v = r\omega$, $r = 50m$

$v = 50 \cdot 0{,}0628$

$v = 3{,}14\ m/s$



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Todos los marcadores

Exercise 1: The Ceiling Fan (Frequency & Velocity)

- **Scenario:** A ceiling fan blade (radius $r = 0.6$ m) rotates at 300 RPM (revolutions per minute).
- **Tasks:**
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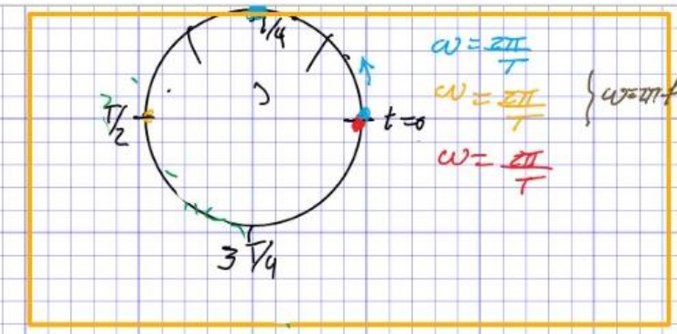

1. $RPM = 300\ rev/min = 300\ rev/60s = 5\ rev/s \Rightarrow \boxed{f = 5\ Hz\ (Hertz)}$

$T = \frac{1}{f} \Rightarrow \boxed{T = \frac{1}{5} = 0{,}2\ s}$

T: periodo: tiempo tarda 1 vuelta

1 vuelta → 0,2 s
10 vueltas → 2 s
100 " → 20 s

2. velocidad angular ω (rad/s)

$\omega = \frac{\Delta\theta}{\Delta t} = 2\pi f = 2\pi \cdot 5 = 10\pi \simeq 31{,}4$

$= 10 \cdot 3{,}1415921\ldots$

$= 31{,}4159\ldots$

$\boxed{\omega = 10\pi\ rad/s}$

3. rapidez lineal $\boxed{v = r\omega = 0{,}6 \cdot 10\pi = 6\pi\ m/s}$

4. $\boxed{v = r\omega = 0{,}3 \cdot 10\pi = 3\pi\ m/s}$

Degree Measure	Radian Measure
30°	$\frac{\pi}{6}$
60°	$\frac{\pi}{3}$
90°	$\frac{\pi}{2}$
120°	$\frac{2\pi}{3}$
135°	$\frac{3\pi}{4}$
180°	π

TAREA.

Example 6.1.1: How Fast Does the Car Tire Spin?

Calculate the angular velocity of a 0.300 m radius car tire when the car travels at $15.0 m/s$ (about $54\ km/h$). See Figure.

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Todos los marcadores

Example 6.1.1: How Fast Does the Car Tire Spin?

Calculate the angular velocity of a 0.300 m radius car tire when the car travels at $15.0 m/s$ (about $54\ km/h$). See Figure.

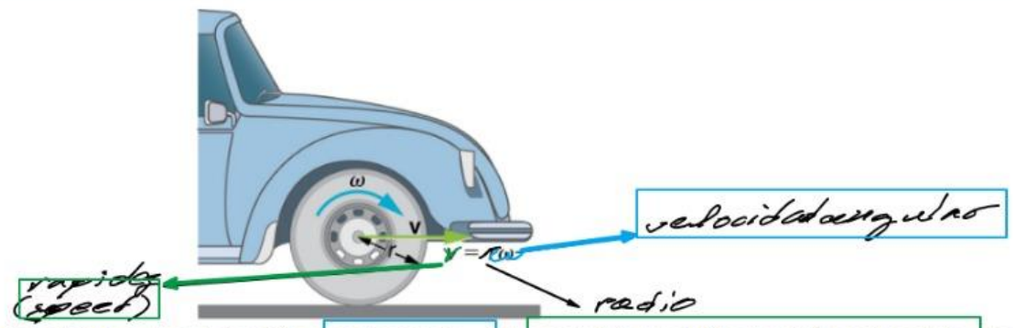


Figure 6.1.4: A car moving at a velocity v to the right has a tire rotating with an angular velocity ω. The speed of the tread of the tire relative to the axle is v, the same as if the car were jacked up. Thus the car moves forward at linear velocity $v = r\omega$, where r is the tire radius. A larger angular velocity for the tire means a greater velocity for the car.

Rueda fotomo

$v = r\omega = 0{,}3 \cdot \omega = 15$

$\omega = \frac{15}{0{,}3}$

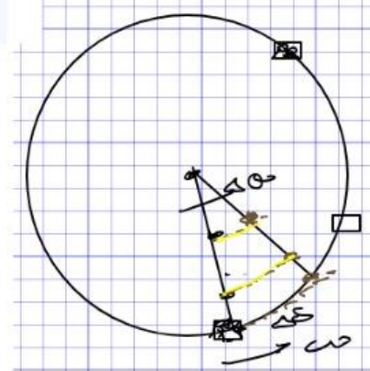
$\omega = \frac{15 \cdot 10}{0{,}3 \cdot 10} = \frac{150}{3} = 50\ rad/s.$

$\frac{4/2}{6/2} = \frac{2}{3}$

$\omega = \frac{2\pi}{120\ s.}$

$\omega = \frac{\pi}{60}\ rad/s$

Δ : variación cambio



360° (vuelta) 2π rad.

$\Delta s = r\Delta\theta$

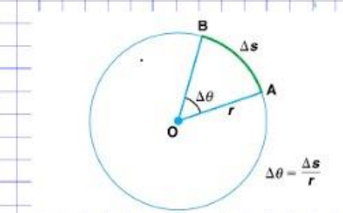
r: radio

Δs: variación longitud arco

Δθ: variación angular

$\omega = \frac{\Delta\theta}{\Delta t}$, Δθ: delta theta

Δt: delta t



$\frac{\Delta s}{\Delta t} = \frac{r\Delta\theta}{\Delta t}$

$v\left(\frac{m}{s}\right) = r\omega\ \left(\frac{m}{s}\right)$

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